

Technical Report: Dataset-Prior Residual Prompt Adaptation for Robust Cross-Dataset Transfer

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Phase 1 Technical Summary

Abstract

This report summarizes the current stage of a research project on dataset-level prior guided prompt learning for CLIP-based visual recognition. The project builds upon CoOp and proposes a dataset-conditioned residual prompt adapter, named CoOpPriorRes. The current main version, Safe PriorRes, injects dataset-level task priors through an identity-centered residual formulation. The central empirical finding is that the method is not primarily an in-domain accuracy booster, but a robustness-oriented prompt adaptation mechanism for cross-dataset transfer. In Phase 1, the project completed cross-dataset domain generalization (DG), dataset-prior ablation, base-to-new (B2N) auxiliary evaluation, residual formulation analysis, and preliminary mechanism-log extraction. The main DG result shows that Safe PriorRes improves over CoOp by +1.28 points on average across three source datasets, while the non-identity Legacy PriorRes variant is unstable and obtains -0.82 on average. The Safe-Real / Safe-Mean / Safe-Shuffle ablation further shows that the gains come from source-aligned dataset priors rather than merely from additional residual adapter parameters.

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1 Executive Summary

One-paragraph architecture summary. The project proposes a dataset-prior residual prompt adapter for CoOp-style prompt learning. For a source dataset $\mathcal{D}$, a compact task-level feature vector $\mathbf{z}_{\mathcal{D}}$ is first extracted from dataset statistics such as class count, semantic similarity, intra-class dispersion, inter-class dispersion, and a Fisher-like separability ratio. A lightweight prior adapter maps $\mathbf{z}_{\mathcal{D}}$ to an initial context-token gate $\mathbf{a}_0$. During prompt learning, the model learns a deviation $\Delta\mathbf{a}$ and forms $\mathbf{a} = \mathbf{a}_0 + \Delta\mathbf{a}$. Instead of directly replacing the CoOp context tokens, the method applies an identity-centered residual update

$$\mathbf{C}_{\text{eff}} = \mathbf{C} + \lambda_t(\mathbf{a} - \mathbf{a}_0) \odot \mathbf{U},$$

where $\mathbf{C}$ is the original learnable CoOp context, $\mathbf{U}$ is a learnable residual direction, and λ_t is a warmup/ramp coefficient. Because $\mathbf{a} = \mathbf{a}_0$ at initialization, the effective prompt starts exactly from vanilla CoOp. This is the main difference from the legacy residual form $\mathbf{C} + \lambda_t(\mathbf{a} - 1) \odot \mathbf{U}$, which can inject a non-zero perturbation at initialization. Therefore, the method should be understood as a **safe, source-prior-conditioned residual prompt adaptation mechanism** rather than a closed-set dataset-wise fine-tuning trick.

Current status. Phase 1 core experiments are completed. The strongest evidence is cross-dataset DG. B2N and in-domain results are used as auxiliary evidence rather than the main claim. The current paper-level story is:

1. CoOp-style prompt learning does not explicitly model dataset-level distributional properties.
2. Dataset-level priors can be injected into prompt learning through a residual adapter.
3. Naive residual injection is not identity-preserving and can be unstable.
4. Safe PriorRes uses identity-centered residual adaptation and is more robust across source datasets.
5. Real / Mean / Shuffle ablations show that source-aligned dataset priors are necessary.

2 Motivation and Inspiration

2.1 Background: Prompt Learning for CLIP

CLIP-style vision-language models provide a strong open-vocabulary recognition backbone. Prompt learning methods such as CoOp replace manually designed text prompts with learnable continuous context tokens while keeping the CLIP image and text encoders frozen. A typical CoOp prompt for class y can be written as

$$\mathbf{p}_y = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m, \text{name}(y)],$$

where $\mathbf{c}_1, \dots, \mathbf{c}_m$ are learnable context tokens and m is the number of context tokens.

CoOp is parameter-efficient, but it learns a static prompt for a dataset and does not explicitly represent the distributional properties of that dataset. Different datasets may have very different class granularity, visual diversity, semantic relatedness, and domain characteristics. For example, a scene-centric source such as SUN397 differs substantially from fine-grained object datasets such as OxfordPets or StanfordCars, and remote-sensing datasets such as EuroSAT introduce an even larger domain gap.

2.2 Core Research Question

The project asks:

Can dataset-level task priors be safely injected into prompt learning to improve robustness under dataset shift?

The emphasis is on **safe injection**. If a dataset prior directly perturbs prompts at initialization, the model may deviate from the well-established CoOp baseline before learning whether the deviation is useful. This motivates an identity-centered residual design.

2.3 Clarifying the Main Task: DG rather than Closed-Set Target Adaptation

A key clarification is that the main task is **cross-dataset domain generalization**. In this setting:

1. A model is trained on a source dataset $\mathcal{D}_s$.
2. The dataset prior $\mathbf{z}_{\mathcal{D}_s}$ is computed only from the source dataset.
3. After training, all model weights are frozen.
4. The model is directly evaluated on each target dataset $\mathcal{D}_t$.
5. No target dataset prior is computed or injected during target evaluation.

Thus, the main DG setting does **not** assume access to target test-set statistics. This is different from a closed-set target adaptation setting. The method is better understood as learning a source-conditioned prompt representation that generalizes better to unseen target distributions.

3 Dataset-Level Task Prior

3.1 Notation

Let a dataset be

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

with C classes. Let $\mathcal{I}_c = \{i : y_i = c\}$ be the index set of samples belonging to class c , and $n_c = |\mathcal{I}_c|$.

Let $E_v(\cdot)$ be the frozen CLIP image encoder and $E_t(\cdot)$ be the frozen CLIP text encoder. Image and text embeddings are normalized:

$$\mathbf{v}_i = \frac{E_v(x_i)}{\|E_v(x_i)\|_2}, \quad \mathbf{t}_c = \frac{E_t(\text{name}(c))}{\|E_t(\text{name}(c))\|_2}.$$

The class visual prototype is

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} \mathbf{v}_i, \quad \hat{\boldsymbol{\mu}}_c = \frac{\boldsymbol{\mu}_c}{\|\boldsymbol{\mu}_c\|_2}.$$

3.2 Extracted Dataset Features

Class-count feature. The number of classes is

$$C_{\mathcal{D}} = |\mathcal{Y}_{\mathcal{D}}|,$$

and the transformed feature uses

$$f_C = \log(C_{\mathcal{D}} + \epsilon).$$

Semantic similarity. Semantic similarity measures how close class names are in CLIP text space:

$$S_{\text{sem}}(\mathcal{D}) = \frac{2}{C(C-1)} \sum_{1 \leq c < c' \leq C} \cos(\mathbf{t}_c, \mathbf{t}_{c'}).$$

A larger value indicates that class names are semantically closer, which may imply a more fine-grained semantic task.

Intra-class dispersion. Intra-class dispersion measures how visually diverse samples are inside each class:

$$D_{\text{intra}}(\mathcal{D}) = \frac{1}{C} \sum_{c=1}^C \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} (1 - \cos(\mathbf{v}_i, \hat{\boldsymbol{\mu}}_c)).$$

The transformed version is

$$f_{\text{intra}} = \log(D_{\text{intra}} + \epsilon).$$

Inter-class dispersion. Inter-class dispersion measures how far visual class prototypes are from one another:

$$D_{\text{inter}}(\mathcal{D}) = \frac{2}{C(C-1)} \sum_{1 \leq c < c' \leq C} (1 - \cos(\hat{\boldsymbol{\mu}}_c, \hat{\boldsymbol{\mu}}_{c'})).$$

The transformed version is

$$f_{\text{inter}} = \log(D_{\text{inter}} + \epsilon).$$

Fisher-like separability. A Fisher-like ratio compares inter-class separation with intra-class variation:

$$R_{\text{Fisher}}(\mathcal{D}) = \frac{D_{\text{inter}}(\mathcal{D})}{D_{\text{intra}}(\mathcal{D}) + \epsilon}.$$

A larger value indicates that classes are relatively easier to separate in the CLIP embedding space. The transformed version is

$$f_{\text{Fisher}} = \log(R_{\text{Fisher}} + \epsilon).$$

Final task feature. The transformed task feature is

$$\mathbf{z}_{\mathcal{D}} = [f_C, S_{\text{sem}}, f_{\text{intra}}, f_{\text{inter}}, f_{\text{Fisher}}].$$

In implementation, the exact serialized feature values are saved in

`outputs/task_features/<dataset>_train.json`.

3.3 Why VLM-Based Features Are Reasonable but Limited

The task features depend on a frozen vision-language encoder, currently CLIP RN50. This dependency is reasonable because the measured quantities, such as semantic similarity and visual separability, are naturally defined in a representation space. However, these statistics are not absolute dataset properties independent of the encoder. They should be interpreted as **CLIP-space task priors**. This is a limitation but not a fatal flaw: the goal is not to estimate universal dataset difficulty, but to provide the prompt learner with a compact summary of how the source dataset appears to the frozen CLIP backbone.

4 Method: CoOpPriorRes

4.1 CoOp Baseline

For an input image x and class c , CoOp constructs a learnable text prompt with context tokens $\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_m]$:

$$\mathbf{p}_c = [\mathbf{c}_1, \dots, \mathbf{c}_m, \text{name}(c)].$$

The CLIP text encoder gives a class text feature

$$\mathbf{w}_c = T(\mathbf{p}_c),$$

and the image encoder gives

$$\mathbf{u} = V(x).$$

The classification probability is computed by cosine similarity:

$$P(y = c \mid x) = \frac{\exp(\cos(\mathbf{u}, \mathbf{w}_c)/\tau)}{\sum_{j=1}^C \exp(\cos(\mathbf{u}, \mathbf{w}_j)/\tau)}.$$

The training objective is the standard cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{B}|} \sum_{(x_i, y_i) \in \mathcal{B}} \log P(y_i \mid x_i).$$

4.2 Prior Adapter

The dataset feature $\mathbf{z}_{\mathcal{D}}$ is fed into a lightweight adapter:

$$\mathbf{a}_0 = f_{\theta}(\mathbf{z}_{\mathcal{D}}), \quad \mathbf{a}_0 \in \mathbb{R}^m,$$

where $m = 16$ in the current experiments.

During training, a learnable deviation is introduced:

$$\mathbf{a} = \mathbf{a}_0 + \Delta\mathbf{a}.$$

The residual direction is

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m] \in \mathbb{R}^{m \times d},$$

where d is the CLIP text embedding dimension.

4.3 Safe Residual Formulation

The current main method uses the identity-centered residual:

$$\mathbf{C}_{\text{safe}} = \mathbf{C} + \lambda_t(\mathbf{a} - \mathbf{a}_0) \odot \mathbf{U}.$$

Here $\odot$ means token-wise modulation: each scalar gate difference $(a_j - a_{0,j})$ modulates the corresponding residual direction $\mathbf{u}_j$.

At initialization,

$$\mathbf{a} = \mathbf{a}_0 \quad \Rightarrow \quad \mathbf{C}_{\text{safe}} = \mathbf{C}.$$

Therefore, the model starts exactly from vanilla CoOp.

4.4 Residual Schedule

The residual strength λ_t is controlled by a warmup/ramp schedule:

$$\lambda_t = \lambda_{\max} \cdot \min \left(1, \max \left(0, \frac{t - T_{\text{warm}}}{T_{\text{ramp}}} \right) \right),$$

where $T_{\text{warm}} = 5$, $T_{\text{ramp}} = 10$, and $\lambda_{\max} = 1.0$ in the current configuration.

4.5 Legacy Residual Formulation

The legacy variant is

$$\mathbf{C}_{\text{legacy}} = \mathbf{C} + \lambda_t(\mathbf{a} - \mathbf{1}) \odot \mathbf{U}.$$

At initialization, $\mathbf{a} = \mathbf{a}_0$, so

$$\mathbf{C}_{\text{legacy}} = \mathbf{C} + \lambda_t(\mathbf{a}_0 - \mathbf{1}) \odot \mathbf{U}.$$

If $\mathbf{a}_0 \neq \mathbf{1}$, this formulation injects a non-zero perturbation before training learns whether the perturbation is useful. In the experiments, this formulation can be stronger for certain source datasets but is less stable across sources.

4.6 Training Objective

The current main model disables the sample-weighting b branch and uses only the context-side a branch. The optimization objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \|\Delta\mathbf{a}\|_2^2,$$

where α is a small regularization coefficient. The CLIP image and text encoders remain frozen.

5 Experimental Protocol

5.1 Default Configuration

Table 1: Default experimental configuration.

Item	Setting
Backbone	CLIP RN50
Trainer	CoOp / CoOpPriorRes
Context length m	16
Shots k	16
Class-specific context	False
Class token position	End
Seeds	1, 2, 3
Precision	fp16
Train batch size	32
Test batch size	100
Main PriorRes setting	Safe no_b

The current main configuration is:

USE_B=False, USE_CONTEXT_GATING=True, USE_LEGACY_RESIDUAL=False.

The training uses no alternate optimization:

ALTERNATE_OPT=False, META_LR_RATIO = 0.3.

5.2 Cross-Dataset DG Protocol

Cross-dataset DG is the main experimental setting. It uses a strict source-only protocol:

1. Train the model on a source dataset $\mathcal{D}_s$.
2. Compute the task prior only from $\mathcal{D}_s$.
3. Freeze the trained model.
4. Evaluate it directly on each target dataset $\mathcal{D}_t$.
5. Do not compute or inject any target dataset prior during target evaluation.

This protocol directly addresses the concern that dataset-wise priors might leak target-set information. In the main DG setting, only the source prior is used.

5.3 Base-to-New Protocol

B2N evaluates whether a prompt learned on base classes generalizes to new classes within the same dataset. The model is trained on base classes and evaluated on both base and new classes. The harmonic mean is

$$\text{HM} = \frac{2 \cdot \text{Acc}_{\text{base}} \cdot \text{Acc}_{\text{new}}}{\text{Acc}_{\text{base}} + \text{Acc}_{\text{new}}}.$$

The current B2N result is auxiliary. In the current version, the prior is dataset-level rather than base-class-only. Therefore, B2N should not be used as the main evidence of strict open-class generalization. It is used to check whether the method severely damages base-to-new behavior.

5.4 Safe Prior Ablation Protocol

Three prior settings are compared:

1. **Safe-Real**: use the true source dataset feature.
2. **Safe-Mean**: use a global averaged feature.
3. **Safe-Shuffle**: use a fixed mismatched dataset feature.

The shuffle mapping is:

$$\text{Caltech101} \rightarrow \text{Food101}, \quad \text{Food101} \rightarrow \text{SUN397}, \quad \text{SUN397} \rightarrow \text{Caltech101}.$$

Safe-Mean tests whether the adapter alone is sufficient. Safe-Shuffle tests whether any real dataset prior is sufficient, or whether the prior must be aligned with the source dataset.

6 Main Results

6.1 Cross-Dataset DG: Source-Level Pivot Table

Table 2: Cross-dataset DG source-level summary. Values are average accuracy deltas over target datasets.

Source	Safe-CoOp	Legacy-CoOp	Legacy-Safe
Caltech101	+1.43	+2.24	+0.81
Food101	+2.21	-0.04	-2.24
SUN397	+0.20	-4.65	-4.85
Overall	+1.28	-0.82	-2.10

Interpretation. Safe PriorRes is more robust across source datasets than Legacy PriorRes. Legacy is stronger on Caltech101, but it is unstable on Food101 and severely negative on SUN397. Therefore, Legacy should be interpreted as an aggressive residual variant, while Safe is the conservative and robust version.

6.2 Cross-Dataset DG: Target-Level Detailed Table

Table 3: Detailed cross-dataset DG results.

Source	Target	CoOp	Safe	Legacy	Safe-CoOp	Legacy-CoOp	Legacy-Safe
Caltech101	OxfordPets	73.33	73.43	72.67	+0.10	-0.67	-0.77
Caltech101	EuroSAT	16.17	24.50	26.13	+8.33	+9.97	+1.63
Caltech101	DTD	29.33	30.63	31.30	+1.30	+1.97	+0.67
Caltech101	Food101	65.43	67.00	68.13	+1.57	+2.70	+1.13
Caltech101	OxfordFlowers	49.10	49.47	49.83	+0.37	+0.73	+0.37
Caltech101	StanfordCars	49.10	49.27	50.20	+0.17	+1.10	+0.93
Caltech101	FGVCAircraft	11.93	10.87	11.67	-1.07	-0.27	+0.80
Caltech101	UCF101	48.10	48.37	50.23	+0.27	+2.13	+1.87
Caltech101	SUN397	47.77	49.60	50.27	+1.83	+2.50	+0.67
Food101	OxfordPets	55.00	61.10	44.47	+6.10	-10.53	-16.63
Food101	EuroSAT	17.40	26.67	22.90	+9.27	+5.50	-3.77
Food101	DTD	20.63	22.57	21.07	+1.93	+0.43	-1.50
Food101	OxfordFlowers	40.87	40.97	41.63	+0.10	+0.77	+0.67
Food101	Caltech101	65.80	65.77	66.60	-0.03	+0.80	+0.83
Food101	StanfordCars	49.37	48.87	49.73	-0.50	+0.37	+0.87
Food101	FGVCAircraft	9.47	7.53	5.57	-1.93	-3.90	-1.97

Source	Target	CoOp	Safe	Legacy	Safe-CoOp	Legacy-CoOp	Legacy-Safe
Food101	UCF101	46.17	46.90	47.40	+0.73	+1.23	+0.50
Food101	SUN397	34.13	38.33	39.13	+4.20	+5.00	+0.80
SUN397	OxfordPets	65.60	65.17	54.87	-0.43	-10.73	-10.30
SUN397	EuroSAT	25.33	20.30	24.33	-5.03	-1.00	+4.03
SUN397	DTD	25.83	30.40	24.13	+4.57	-1.70	-6.27
SUN397	Food101	63.00	66.40	61.23	+3.40	-1.77	-5.17
SUN397	OxfordFlowers	46.97	45.77	40.33	-1.20	-6.63	-5.43
SUN397	Caltech101	77.73	78.80	74.70	+1.07	-3.03	-4.10
SUN397	StanfordCars	52.07	52.70	46.07	+0.63	-6.00	-6.63
SUN397	FGVCAircraft	10.20	10.43	7.30	+0.23	-2.90	-3.13
SUN397	UCF101	53.50	52.07	45.40	-1.43	-8.10	-6.67

6.3 Safe Dataset-Prior Ablation

Table 4: Safe prior ablation: source-level summary.

Source	Real-CoOp	Mean-CoOp	Shuffle-CoOp	Real-Mean	Real-Shuffle
Caltech101	+1.43	+1.10	+1.10	+0.33	+0.33
Food101	+2.21	+1.57	+1.29	+0.63	+0.92
SUN397	+0.20	-2.76	-3.97	+2.96	+4.17
Overall	+1.28	-0.03	-0.53	+1.31	+1.81

Interpretation. Safe-Real is the only variant that gives a clear positive overall gain. Safe-Mean almost degenerates to CoOp-level performance, and Safe-Shuffle becomes negative. This shows that the improvement is not mainly caused by adding a residual adapter. The source-aligned dataset prior is necessary.

SUN397 case. SUN397 provides the strongest evidence:

$$\text{Safe-Real} = +0.20, \quad \text{Safe-Mean} = -2.76, \quad \text{Safe-Shuffle} = -3.97.$$

Here the real prior does not produce a large positive gain, but it prevents the large negative transfer caused by generic or mismatched priors. This is important because SUN397 is scene-centric and has a different domain structure from many object-centric target datasets.

6.4 Base-to-New Results

Table 5: B2N delta summary: PriorRes minus CoOp.

Dataset	Δ_{Base}	Δ_{New}	Δ_{HM}
Caltech101	+0.07	-3.23	-1.72
OxfordPets	+0.00	-0.90	-0.43
DTD	+0.27	+9.13	+7.99
EuroSAT	-0.40	-1.37	-2.01
Food101	-0.27	+0.43	+0.11
OxfordFlowers	-0.53	-4.07	-3.14
StanfordCars	+0.43	-1.83	-0.96
FGVCAircraft	-0.40	+1.97	+1.21
UCF101	-0.67	+2.27	+1.30
SUN397	-0.20	-1.87	-1.18
Average	-0.17	+0.05	+0.12

Interpretation. B2N is not the main strength of the method. On average, PriorRes is comparable to CoOp. Strong positive cases appear on DTD, FGVC-Aircraft, and UCF101, but the gain is not universal. Moreover, the current B2N prior is dataset-level rather than base-class-only, so B2N should be presented as auxiliary evidence rather than strict proof of open-class generalization.

6.5 In-Domain and Residual Formulation Summary

The in-domain experiments are auxiliary. The correct interpretation is:

Safe PriorRes preserves CoOp-level in-domain performance, but is not designed as a universal in-domain accu

The residual formulation analysis suggests:

- In-domain residual ablation: Safe-CoOp average +0.59, Safe-Legacy average +0.40.
- DG residual interpretation: Legacy can be stronger for Caltech101-source transfer but is not robust across sources.
- Therefore, the key message is a **safety-transfer trade-off**.

7 Mechanism Analysis: Current Evidence and Next Step

7.1 What Has Been Logged

The mechanism logs currently contain tracked quantities such as:

$$m_{\text{eff}}, \quad k_{\text{eff}}, \quad a_{0,\text{mean}}, \quad b_{0,\text{mean}}, \quad \lambda_t, \quad \Delta a.$$

They are saved in files such as:

`safe_main_train_meff_keff_a0.txt`, `safe_mean_train_meff_keff_a0.txt`, `safe_shuffle_train_meff_keff_a0.txt`

These logs are enough to support a first-stage qualitative mechanism discussion, but a complete mechanism figure still needs to be prepared.

7.2 Current Mechanistic Interpretation

Safe residual starts from identity. By construction,

$$\mathbf{C}_{\text{safe}}(t=0) = \mathbf{C}.$$

Therefore, any early performance change must come after the residual branch becomes active through λ_t and learned $\Delta \mathbf{a}$. This explains why Safe is more stable across source datasets.

Legacy residual is aggressive. The legacy form can inject

$$\lambda_t(\mathbf{a}_0 - \mathbf{1}) \odot \mathbf{U}$$

before a meaningful deviation is learned. This can act as a useful transfer perturbation for Caltech101, where Legacy obtains +2.24 over CoOp, but it becomes unstable for SUN397, where Legacy obtains −4.65.

Real prior prevents negative transfer. The Safe-Real / Mean / Shuffle results show that real source priors are especially useful for avoiding negative transfer. This is clearest on SUN397:

$$\text{Real-CoOp} = +0.20, \quad \text{Mean-CoOp} = -2.76, \quad \text{Shuffle-CoOp} = -3.97.$$

The mechanism hypothesis is that a scene-centric source such as SUN397 requires source-aligned prior modulation; generic or mismatched priors push the prompt in directions that do not transfer well to object-centric targets.

7.3 Recommended Mechanism Figures

For the next version of the report or paper, the following figures should be generated:

1. **Residual norm curve:**

$$\|\lambda_t(\mathbf{a} - \mathbf{a}_0) \odot \mathbf{U}\|_2$$

for Safe-Real, Safe-Mean, Safe-Shuffle, and Legacy.

2. **Gate deviation curve:**

$$\|\Delta \mathbf{a}\|_2 = \|\mathbf{a} - \mathbf{a}_0\|_2.$$

3. **Source-target prior distance vs DG delta:** compare the distance between task features and the observed target accuracy delta.
4. **Case study: SUN397 source** to show why wrong priors and aggressive residuals cause negative transfer.

8 Addressing Potential Concerns

8.1 Does the Method Use Target Dataset Priors in DG?

No. In the main DG setting, the method uses only the source dataset prior during training. At test time, the model is directly evaluated on unseen target datasets without computing target features or injecting target-specific priors. This is the most important clarification for avoiding the misunderstanding that the method uses test-set information.

8.2 Does the Dataset Prior Break Open-Vocabulary Generalization?

In the strict DG setting, the prior summarizes the source dataset, not the target test set. Therefore, it does not require knowing the target distribution. The method is closer to learning a better source-conditioned prompt representation than to closed-set target adaptation.

8.3 Are the Dataset Features Heuristic?

Yes, the current features are partly heuristic. They are CLIP-space estimates of semantic similarity, intra-class dispersion, inter-class dispersion, and separability. The current contribution is empirical and methodological rather than a complete theory of dataset difficulty. However, the Real / Mean / Shuffle ablation provides evidence that these features are not arbitrary: real source priors outperform mean and mismatched priors.

8.4 Is B2N Fully Rigorous?

The current B2N setup should be treated carefully because the prior is dataset-level rather than base-class-only. Therefore, B2N is auxiliary. The main claim relies on strict source-only DG, where no target priors are used.

8.5 Deployment Interpretation

In a deployment scenario, the method can be trained on a known support/source dataset and then applied to new images or target domains without recomputing a target prior. This is consistent with the DG protocol. The method does not require collecting a full target dataset at test time.

9 Limitations and Future Work

1. **Mechanism explanation is still incomplete.** Current logs provide a starting point, but residual norm and gate dynamics should be visualized.
2. **The method is currently CoOp-based.** To strengthen publication potential, the adapter should be tested on CoCoOp or CoCoOpS.
3. **B2N needs a stricter base-only prior version.** The current B2N result is auxiliary.
4. **Dataset features depend on CLIP space.** Future work should test encoder sensitivity or alternative feature extractors.
5. **ImageNet scaling remains optional.** It may improve completeness but is not required for the current Phase 1 conclusion.

10 Conclusion

This project proposes Safe PriorRes, an identity-centered dataset-prior residual prompt adapter built upon CoOp. The method extracts dataset-level task priors and uses them to modulate prompt context tokens through a safe residual branch. The strongest empirical evidence appears in cross-dataset DG: Safe PriorRes obtains an overall +1.28 improvement over CoOp across three source datasets, while Legacy PriorRes obtains -0.82 and shows source-dependent instability. The Safe-Real / Mean / Shuffle ablation further confirms that source-aligned dataset priors, not merely additional adapter parameters, are responsible for the main gains. B2N results are comparable on average and should be used as auxiliary evidence. The next important step is to turn the mechanism logs into residual-dynamics visualizations and to test whether the adapter generalizes beyond CoOp, especially on CoCoOp or CoCoOpS.